Report: Financial Risk Analysis and Customer Behavior Insights Using Python.

1. Introduction

This project focuses on analyzing financial transaction data from Goldman Sachs to understand customer behavior, transaction patterns, and financial risks. With the rapid growth of digital banking, it has become essential for financial institutions to monitor customer activity and identify potential risks in real-time.

The purpose of this analysis is to explore how customers interact with their accounts, identify trends in debit and credit transactions, and detect abnormal financial behavior. The insights derived from this project can help improve customer engagement strategies and enhance risk management systems.

2. Objective

The main objective of this project is to perform a comprehensive analysis of customer financial behavior and risk using Python. The key goals include:

- Cleaning and preprocessing the dataset to ensure accuracy
- Analyzing transaction trends over time
- Segmenting customers based on activity levels
- Identifying financial risks such as anomalies and irregular transactions
- Conducting hypothesis testing to validate assumptions

This analysis aims to provide actionable insights that support better decision-making for financial institutions.

3. Dataset Description

The dataset used in this project contains transactional data of customer accounts from Goldman Sachs. It includes the following key attributes:

- Account ID
- Customer ID
- Transaction Date
- Transaction Type (Credit/Deposit, Debit/Withdrawal)
- Transaction Amount
- Account Balance
- Risk Score

- Credit Rating

This dataset provides a comprehensive view of customer financial activity, enabling detailed analysis of transaction behavior and risk patterns.

4. Data Cleaning and Preparation

The dataset underwent a thorough cleaning and preprocessing process to ensure data quality and consistency.

- The transaction date column was converted into a proper datetime format for time-based analysis.
- Numerical columns such as transaction amount, account balance, risk score, and credit rating were converted into appropriate numeric formats.
- Missing values and invalid entries were removed to maintain data accuracy.
- Categorical fields such as transaction type and account type were standardized for consistency.

These steps ensured that the dataset was clean, structured, and ready for further analysis.

5. Transaction Analysis

A detailed analysis of transaction trends was performed on both monthly and yearly levels.

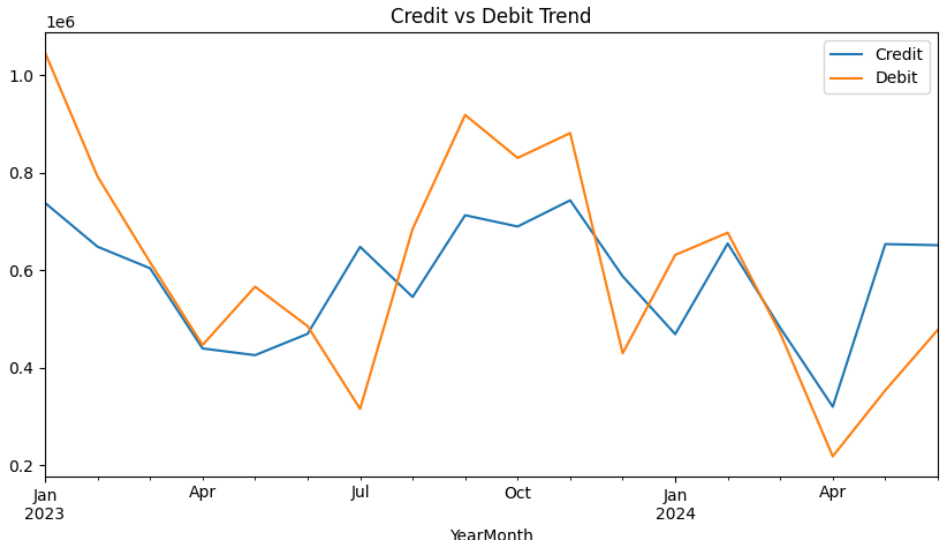
The monthly summary showed fluctuations in credit and debit transactions over time. In the year 2023, debit transactions were consistently higher than credit transactions, resulting in a negative net flow. This indicates higher spending or outflow of funds during that period.

However, in 2024, the trend reversed, showing a positive net flow where credits exceeded debits. This suggests financial recovery and improved inflow of funds.

Top-performing accounts were identified based on high net inflow, while underperforming accounts were identified based on low or negative transaction values.

Dormant accounts were detected by identifying gaps of more than two months between consecutive transactions. A significant number of accounts were found to be inactive, indicating low customer engagement.

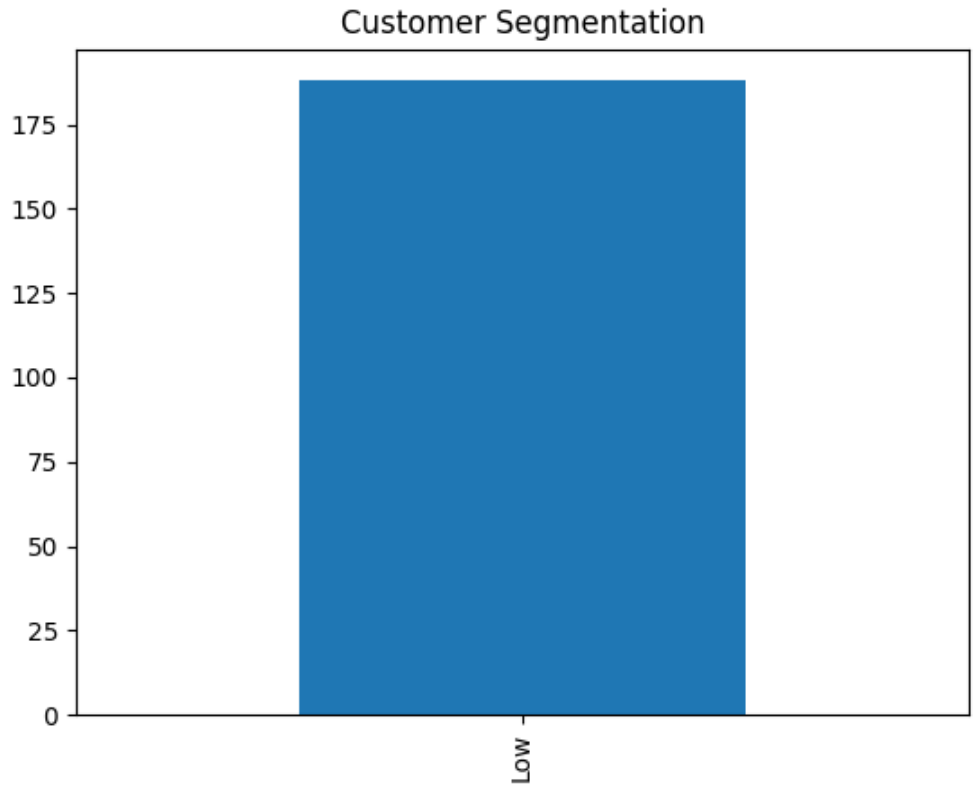
Chart 1: Credit vs Debit :



6. Customer Segmentation

Customers were segmented based on their transaction frequency using the following criteria:

Chart 2: Customer Segmentation



- High Activity: More than 50 transactions
- Medium Activity: Between 20 and 50 transactions
- Low Activity: Less than 20 transactions

The analysis revealed that the majority of customers fall under the low activity segment. This indicates that most users interact minimally with their accounts, which could impact long-term customer retention and engagement.

7. Customer Profiles

Several customer profiles were identified based on transaction patterns and account behavior:

- High Net Inflow Accounts: Customers with significantly higher credit transactions than debit transactions. These accounts are financially strong and contribute positively to the bank.
- High-Frequency Low-Balance Accounts: Customers who perform frequent transactions but maintain low account balances. This behavior may indicate financial instability or high spending patterns.
- Negative or Near-Zero Balance Accounts: Accounts with zero or negative balances, which may indicate overdraft usage or poor financial management.

These profiles help in understanding different customer behaviors and identifying potential risk groups.

8. Financial Risk Analysis

Financial risk was analyzed using multiple methods:

- **Balance Volatility:** Standard deviation was used to measure fluctuations in account balances. High volatility indicates unstable financial behavior.
- **Large Withdrawals:** Accounts with frequent high-value debit transactions were identified as potential risk indicators.
- **IQR Anomaly Detection:** The Interquartile Range (IQR) method was used to detect outliers in transaction amounts. Only a small number of anomalies were detected, indicating overall stability in transaction patterns.

This analysis helps in identifying customers with irregular or suspicious financial activity.

9. Hypothesis Testing

A hypothesis test was conducted to determine whether high-activity customers have higher account balances.

- H0 (Null Hypothesis): High activity customers do not have higher balances
- H1 (Alternative Hypothesis): High activity customers have higher balances

The analysis showed that there is no significant difference between the average balances of high and low activity customers. Therefore, the null hypothesis was not rejected.

This indicates that higher transaction activity does not necessarily imply higher account balances.

10. Key Findings

- Most customers belong to the low activity segment
- A large number of dormant accounts were identified
- Debit transactions exceeded credit transactions in 2023
- Financial recovery was observed in 2024
- Only a small number of anomalies were detected
- High transaction activity does not guarantee higher account balance

11. Recommendations

Based on the analysis, the following recommendations are suggested:

- Improve engagement strategies for dormant and low-activity customers
- Monitor accounts with frequent large withdrawals to reduce risk
- Encourage balanced financial behavior through customer awareness programs
- Implement real-time anomaly detection systems for fraud prevention

These strategies can help financial institutions enhance customer experience and minimize financial risks.

12. Conclusion

This project successfully analyzed customer financial behavior and identified key transaction patterns and risks. The insights gained from this analysis can help financial institutions make data-driven decisions and improve their risk management strategies.

By focusing on customer segmentation, transaction trends, and anomaly detection, banks can better understand customer needs and ensure financial stability.